

Assignment - 2

Solution

1. Simplify:

- a. $(P + Q + R)(P + Q')(P' + R)$
 $= (PP + PQ' + PQ + QQ' + PR + Q'R)(P' + R)$
 $= (P + P(Q' + Q) + PR + Q'R)(P' + R)$
 $= (P(1 + R) + Q'R)(P' + R)$
 $= (P + Q'R)(P' + R)$
 $= PP' + PR + Q'RP' + Q'RR$
 $= PR + Q'R(P' + 1)$
 $= PR + Q'R$
- b. $A'B'C + (A+B+C)' + A'B'C'$
 $= A'B'C + A'B'C' + A'B'C'$
 $= A'B'C + A'B'C'$
 $= A'B'(C + C')$
 $= A'B'(1) = A'B'$
- c. $XYZ + X(Y+Z) + YZ$
 $= XYZ + XY + XZ + YZ$
 $= XY(1+Z) + XZ + YZ$
 $= XY + XZ + YZ$

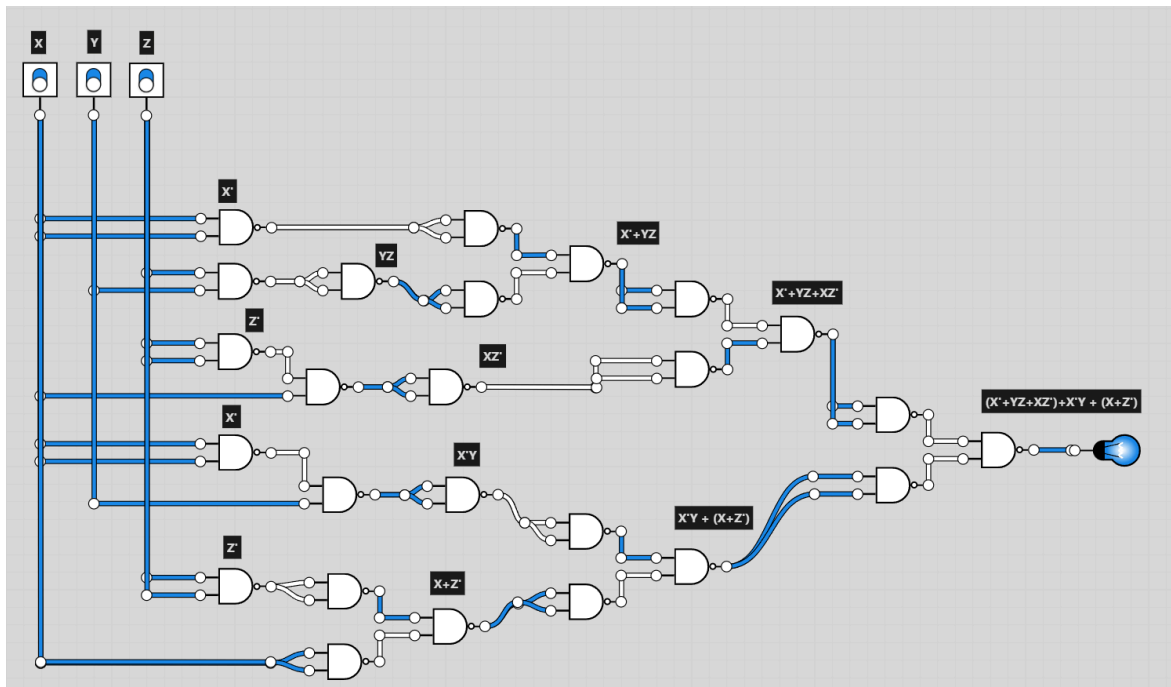
2. Complement:

- a. $A' + C(B + C')(A' + B')(A + C')$
 $= A' \cdot [C + (B \cdot C') + (A' \cdot B') + (A \cdot C')]$
 $= A [C' + (B'C) + (A B) + (A'C)]$
- b. $(P + Q')(P' + R)(Q + R') + S$
 $= (P \cdot Q') + (P' \cdot R) + (Q \cdot R') + S$
 $= (P'Q) + (P R') + (Q'R) \cdot S'$

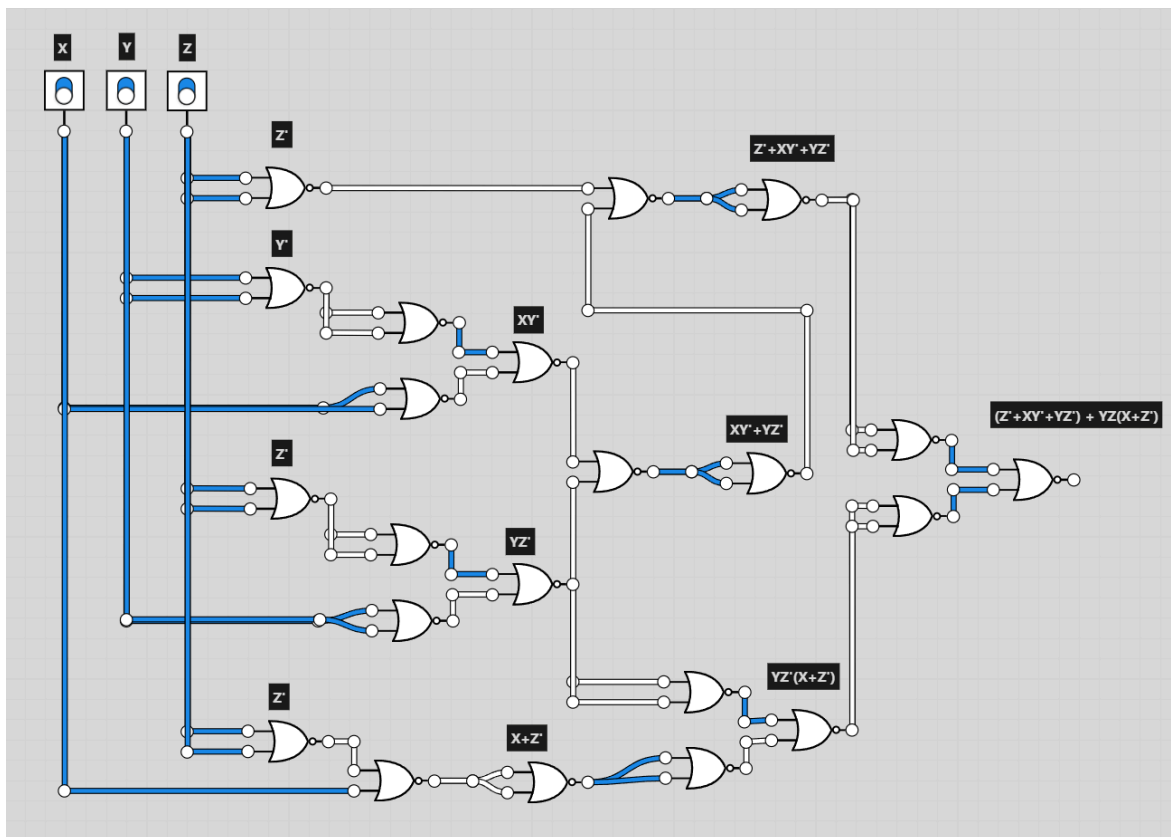
3. SOP: $F = \sum(001, 011, 100, 101, 110)$
 $= A'B'C + A'BC + AB'C' + AB'C + ABC'$

POS: $F = \prod(000, 010, 111)$
 $= (A + B + C)(A + B' + C)(A' + B' + C')$

4. Draw using **NAND gates only**:



5. Draw using **NOR gates only**



6. Find SOP and POS:

SOP:

$$\begin{aligned}F(P, Q, R, S) &= P + QR'S \\&= P(Q+Q')(R+R')(S+S') + (P+P')QR'S \\&= (PQ + PQ')(R+R')(S+S') + PQR'S + P'QR'S \\&= (PQR + PQR' + PQ'R + PQ'R')(S+S') + PQR'S + P'QR'S \\&= PQRS + PQR'S + PQ'RS + PQ'R'S + PQRS' + PQR'S' + PQ'RS' + \\&\quad PQ'R'S' + PQR'S + P'QR'S \\&= PQRS + PQR'S + PQ'RS + PQ'R'S + PQRS' + PQR'S' + PQ'RS' + \\&\quad PQ'R'S' + P'QR'S \\&= \sum (1111, 1101, 1011, 1001, 1110, 1100, 1010, 1000, 0101) \\&= \sum (15, 13, 11, 9, 14, 12, 10, 8, 5) \\&= \sum (5, 8, 9, 10, 11, 12, 13, 14, 15)\end{aligned}$$

POS:

$$\begin{aligned}F(P, Q, R, S) &= P + QR'S \\&= (P+Q)(P+R'S) \\&= (P+Q)(P+R')(P+S) \\&= (P+Q+RR'+SS')(P+QQ'+R'+SS')(P+QQ'+RR'+S) \\&= (P+Q+R+SS')(P+Q+R'+SS')(P+Q+R'+SS')(P+Q'+R'+SS') \\&\quad (P+Q+RR'+S)(P+Q'+RR'+S) \\&= (P+Q+R+S)(P+Q+R+S')(P+Q+R'+S)(P+Q+R'+S')(P+Q+R'+S) \\&\quad (P+Q+R'+S')(P+Q'+R'+S)(P+Q'+R'+S')(P+Q+R+S)(P+Q+R'+S) \\&\quad (P+Q'+R+S)(P+Q'+R'+S) \\&= (P+Q+R+S)(P+Q+R+S')(P+Q+R'+S)(P+Q+R'+S')(P+Q'+R'+S) \\&\quad (P+Q'+R'+S')(P+Q'+R+S) \\&= \Pi(0000, 0001, 0010, 0011, 0110, 0111, 0100) \\&= \Pi(0, 1, 2, 3, 4, 6, 7)\end{aligned}$$